Planet3 Project Proposal

Project Title: Planet3 Gym Membership & Analytics System

Executive Summary

Planet3 is a local fitness center seeking to modernize its membership, subscription, and revenue management processes. Previously operating with paper-based records, the business lacked centralized data storage, performance visibility, and structured reporting capabilities.

This project redesigns Planet3's operations using a normalized SQL Server database and a Power BI analytics layer to support operational efficiency, revenue transparency, and executive decision-making.

Business Problem

Planet3 faced the following operational limitations:

- Manual tracking of memberships and payments
- No centralized system for subscription monitoring
- Limited insight into recurring revenue
- No structured reporting for class performance
- Inability to forecast upcoming payments

These inefficiencies restricted scalability and reduced visibility into financial and operational performance.

Project Objective

The objective of this project is to design and implement a structured database and analytics system that:

- Digitizes membership and subscription management
- Centralizes payment tracking

- Enables recurring revenue monitoring
- Provides executive dashboards for decision support
- Improves data integrity through normalized schema design

System Design & Architect

Backend Database (SQL Server)

The system is built using Microsoft SQL Server and includes:

- Normalized relational schema
- Entity Relationship Diagram (ERD)
- Logical table design
- Primary and foreign key constraints
- T-SQL views optimized for reporting
- Stored procedure to calculate days until next payment

Core Entities:

- Members
- Subscriptions (Normal / VIP)
- Payments
- Classes
- Trainers
- Employees
- Appointments

Business Rules

The system enforces the following rules:

- A member may hold only one active subscription.
- Subscriptions are categorized as Normal or VIP.
- Payments are associated with a subscription and occur monthly.
- Members may register for multiple classes.
- Trainers may instruct multiple classes and serve multiple members.
- Employees report to a supervisor (recursive relationship).
- Equipment must be inspected monthly.

Analytics & Reporting Layer

Power BI dashboards were developed to provide:

- Membership Overview (Total Members, Active Subscriptions)
- Revenue by Subscription Type
- Class Registration Performance
- Next Payment Due Monitoring
- KPI Cards for Executive Review

Reporting was structured using SQL views rather than raw tables to improve clarity, security, and maintainability.

Business Value

This system transforms Planet3 from a manual operation into a data-driven organization by:

- Improving revenue transparency
- Automating reporting processes

- Enabling subscription performance analysis
- Supporting scalable growth through structured data architecture

Project Context

Planet3 was developed as a structured academic case study simulating the modernization of a local gym transitioning from paper-based operations to a centralized digital system.

While the project originated in a team setting, the database implementation, SQL development, reporting views, and Power BI analytics layer were independently designed and built as part of my technical execution.